Linear Algebra

Big Theorems

Vector spaces · linear maps · matrices · spectral theory

~~This five-theorem set spans the core of an undergraduate linear-algebra course. Each theorem is stated informall~~y, argued in prose, and then formalised as a Lean 4 statement typed against Mathlib. Upload this PDF to the Leibniz engine: it scans the statements, formalises the informal text against the encyclopedia, and runs the three gates (Validity, Alignment, Reading) on each theorem.

The theorems span: rank-nullity, invertibility via the determinant, multiplicativity of the determinant, reality of eigenvalues of symmetric matrices, and the Cayley-Hamilton theorem. Each is a landmark result; formal verification of the full proofs requires Mathlib's machinery.

THEOREM 1

1. The Rank-Nullity Theorem

Let f : V -> W be a linear map between finite-dimensional vector spaces. Then dim(range f) + dim(kernel f) = dim V.

Argument. The kernel of f is a subspace of V, and the range (image) is a subspace of W. By the first isomorphism theorem, V / ker f is isomorphic to range f. Taking dimensions: dim V = dim(ker f) + dim(V/ker f) = dim(ker f) + dim(range f). This is the fundamental theorem of linear maps: it constrains how much information a linear transformation can lose.

Consequence. A linear map on a finite-dimensional space is injective iff rank f = dim V (full rank), and surjective onto W iff rank f = dim W.

FORMAL LEAN 4 STATEMENT

```
theorem rank_nullity (K V W : Type*) [Field K] [AddCommGroup V] [Module K V] [AddCommGroup W] [Module K W] [FiniteDimensional K V] [FiniteDimensional K W] (f : V ->_[K] W) : finrank K (LinearMap.range f) + finrank K (LinearMap.ker f) = finrank K V
```

THEOREM 2

2. Invertibility via the Determinant

A square matrix over a field is invertible if and only if its determinant is nonzero.

Argument. The determinant is a multiplicative homomorphism from the matrix ring to the field. If A is invertible with inverse B, then det(A)det(B) = det(I) = 1, so det(A) != 0. Conversely, if det(A) != 0, the classical adjugate formula gives an explicit inverse: A^{-1} = adj(A)/det(A). This bridges algebraic invertibility with a single scalar invariant, and is the theoretical backbone of solving Ax = b via Cramer's rule.

Remark. 'Nonsingular', 'invertible', and 'full rank' coincide for square matrices over a field; each characterisation is useful in different contexts.

FORMAL LEAN 4 STATEMENT

```
theorem invertible_iff_det_ne_zero (K : Type*) [Field K] (A : Matrix (Fin n) (Fin n) K) : Invertible A
<-> det A <> 0
```

THEOREM 3

3. Multiplicativity of the Determinant

*For n x n matrices A and B, det(A * B) = det(A) * det(B).*

Argument. Fix A and consider the map B |-> det(AB)/det(A) (for det(A) != 0). This map is multilinear and alternating in the columns of B, and sends the identity to 1; by the uniqueness of the determinant as such a function, it equals det(B). Hence det(AB) = det(A)det(B). The case det(A) = 0 follows by continuity / polynomial identity. This is the structural reason the determinant detects invertibility: it is a ring homomorphism to the field's multiplicative monoid.

FORMAL LEAN 4 STATEMENT

```
theorem det_mul (K : Type*) [Field K] (A B : Matrix (Fin n) (Fin n) K) : det (A * B) = det A * det B
```

THEOREM 4

4. Eigenvalues of a Real Symmetric Matrix are Real

Every eigenvalue of a real symmetric matrix is a real number.

Argument. Let A be real symmetric (A = A^T) with eigenpair (lambda, v), viewed over C. Compute v* A v = lambda ||v||^2 in two ways. Since A is symmetric and real, v* A v is real, forcing lambda real. Equivalently, a real symmetric matrix is Hermitian, and Hermitian matrices are diagonalisable by a unitary with a real spectrum (the spectral theorem). This is why symmetric matrices govern quadratic forms, second derivatives, and covariance estimation.

Consequence. Real symmetric matrices admit an orthonormal eigenbasis - the foundation of principal component analysis and the spectral decomposition.

FORMAL LEAN 4 STATEMENT

```
theorem real_symmetric_real_eigenvalues (n : Type*) (A : Matrix n n Real) (hA : A.IsSymm) : forall (lam : Complex), A.HasEigenvalue lam -> lam in Set.range ((^) : Real -> Complex)
```

THEOREM 5

5. The Cayley-Hamilton Theorem

Every square matrix satisfies its own characteristic equation: p_A(A) = 0, where p_A(t) = det(tI - A).

Argument. The characteristic polynomial p_A(t) = det(tI - A) is monic of degree n. Cayley-Hamilton asserts that substituting the matrix A for the scalar t annihilates A: p_A(A) = 0. The cleanest proof passes to the algebraic closure: A is triangularisable, and on a triangular form the eigenvalues (roots of p_A) lie on the diagonal, so p_A(A) is strictly upper-triangular nilpotent - hence zero. This underpins the minimal polynomial, the Jordan form, and matrix exponentials in ODE theory.

Significance. Cayley-Hamilton turns a polynomial identity (det(tI - A)) into a matrix identity, linking spectral and algebraic structure.

FORMAL LEAN 4 STATEMENT

```
theorem cayley_hamilton (K : Type*) [Field K] (A : Matrix (Fin n) (Fin n) K) : A.eval (charpoly A) = 0
```

End of big-theorem set. Upload this PDF at leibniz.streamlit.app to scan, formalise, and verify each theorem through the three gates.